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DuoSkin

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MIT Media Lab in collaboration with Microsoft Research*

DuoSkin is a fabrication process that enables anyone to create customized functional devices that can be attached directly on their skin. Using gold metal leaf, a material that is cheap, skin-friendly, and robust for everyday wear, we demonstrate three types of on-skin interfaces: sensing touch input, displaying output, and wireless communication. DuoSkin draws from the aesthetics found in metallic jewelry-like temporary tattoos to create on-skin devices which resemble jewelry. DuoSkin devices enable users to control their mobile devices, display information, and store information on their skin while serving as a statement of personal style. We believe that in the future, on-skin electronics will no longer be black-boxed and mystified; instead, they will converge towards the user friendliness, extensibility, and aesthetics of body decorations, forming a DuoSkin integrated to the extent that it has seemingly disappeared.

Paper to appear at [International Symposium on Wearable Computers 2016 \(ISWC'16\)](#):

Cindy Hsin-Liu Kao, Christian Holz, Asta Roseway, Andres Calvo, Chris Schmandt, "DuoSkin: Rapidly Prototyping On-Skin User Interfaces Using Skin-Friendly Materials.", to appear in ISWC'16. Heidelberg, Germany (September 12-16, 2016). ACM, New York, NY, USA, 8 pages. [\[pdf\]](#)

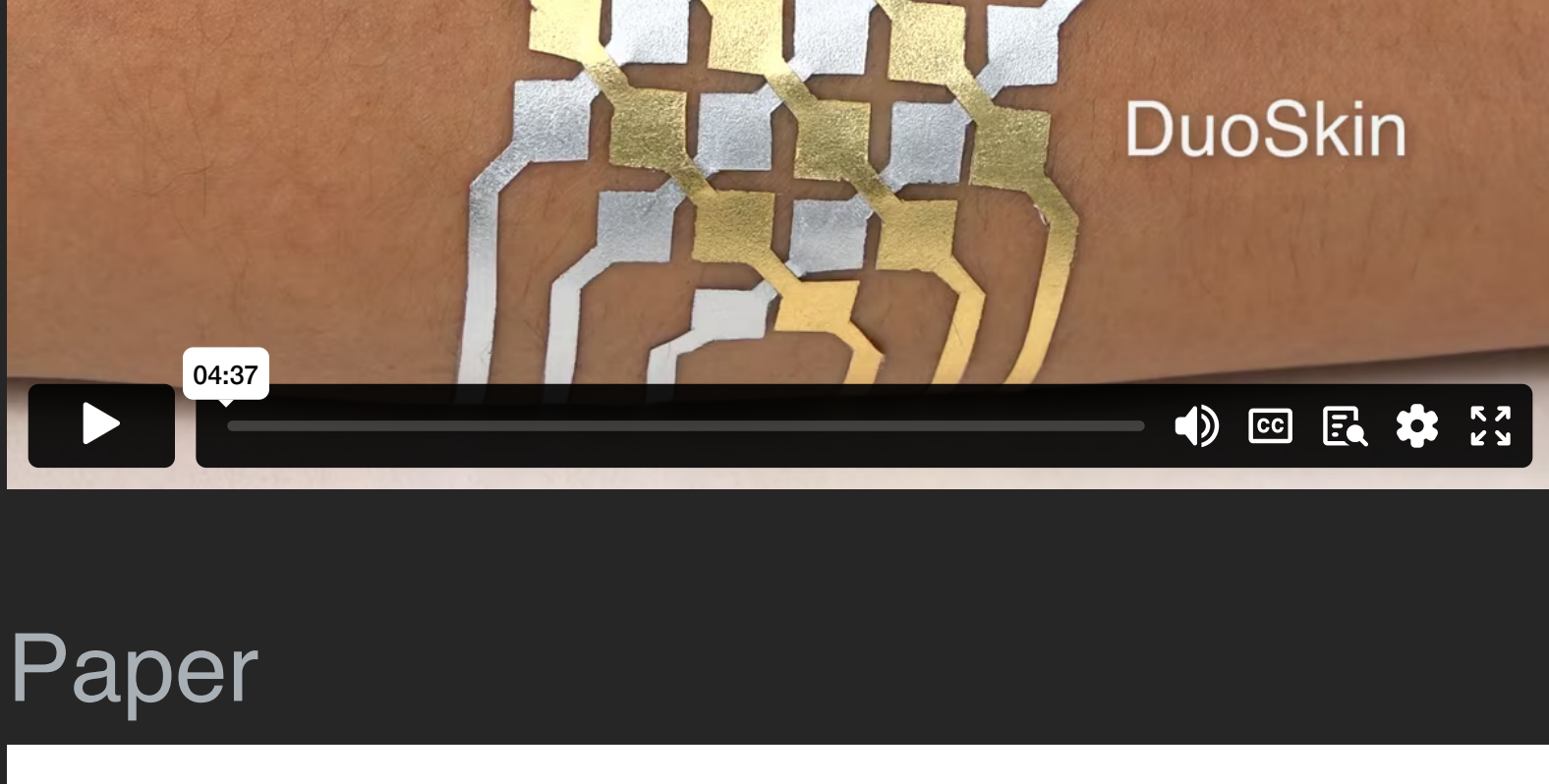
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Watch in Action



Paper

DuoSkin: Rapidly Prototyping On-Skin User Interfaces Using Skin-Friendly Materials

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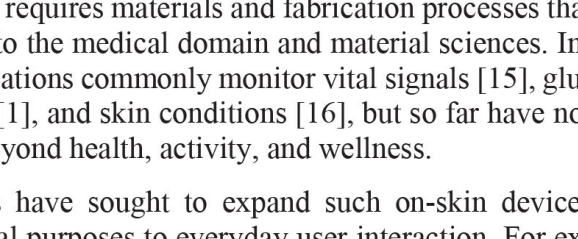
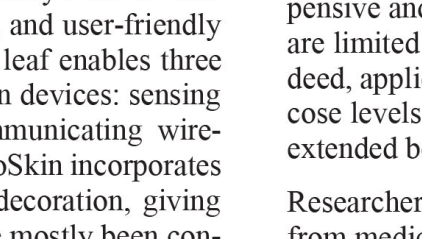
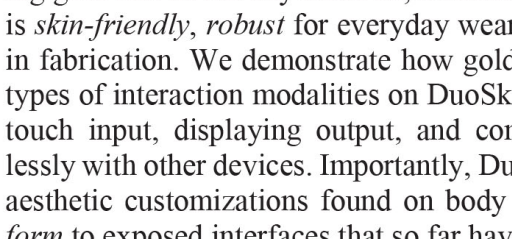


Figure 1: DuoSkin is a rapid prototyping and fabrication process using the skin-friendly material gold leaf to create wearable on-skin user interfaces, such as (a) sensing touch input, (b) displaying the user's mood using a thermochromic tattoo, and (c) communicating and sharing data with other devices through NFC—all while maintaining body decoration aesthetics.

ABSTRACT

Miniature devices have become wearable beyond the form factor of watches or rings—functional devices can now directly affix to the user's skin, unlocking a much wider canvas for electronics. However, building such small and skin-friendly devices currently requires expensive materials and equipment that is mostly found in the medical domain. We present *DuoSkin*, a fabrication process that affords rapidly prototyping functional devices directly on the user's skin using gold leaf as the key material, a commodity material that is *skin-friendly*, *robust* for everyday wear, and user-friendly in fabrication. We demonstrate how gold leaf enables three types of interaction modalities on DuoSkin devices: sensing touch input, displaying output, and communicating wirelessly with other devices. Importantly, DuoSkin incorporates aesthetic customizations found on body decoration, giving *form* to exposed interfaces that so far have mostly been concealed by covers. Our technical evaluation confirmed that gold leaf was more durable and preferable when affixed to skin than current commodity materials during everyday wear. This makes gold leaf a viable material for users to build functional and compelling on-skin devices. In our workshop evaluation, participants were able to customize their own on-skin music controllers that reflected personal aesthetics.

Author Keywords

Electronic skin; Temporary tattoo; Gold leaf; Fabrication; Craft.

ACM Classification Keywords

H.5.m. Information Interfaces and Presentation (e.g. HCI): Miscellaneous

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INTRODUCTION

Everyday devices have now become so small that they can be embedded into fabrics [10], worn as accessories [20], or be attached directly to the user's body on skin. Particularly such epidermal electronics are promising [15,34], as they are always available and expand the sensing modalities of current mobile and wearable devices by sitting directly on skin, which facilitates direct access to users' biomedical signals.

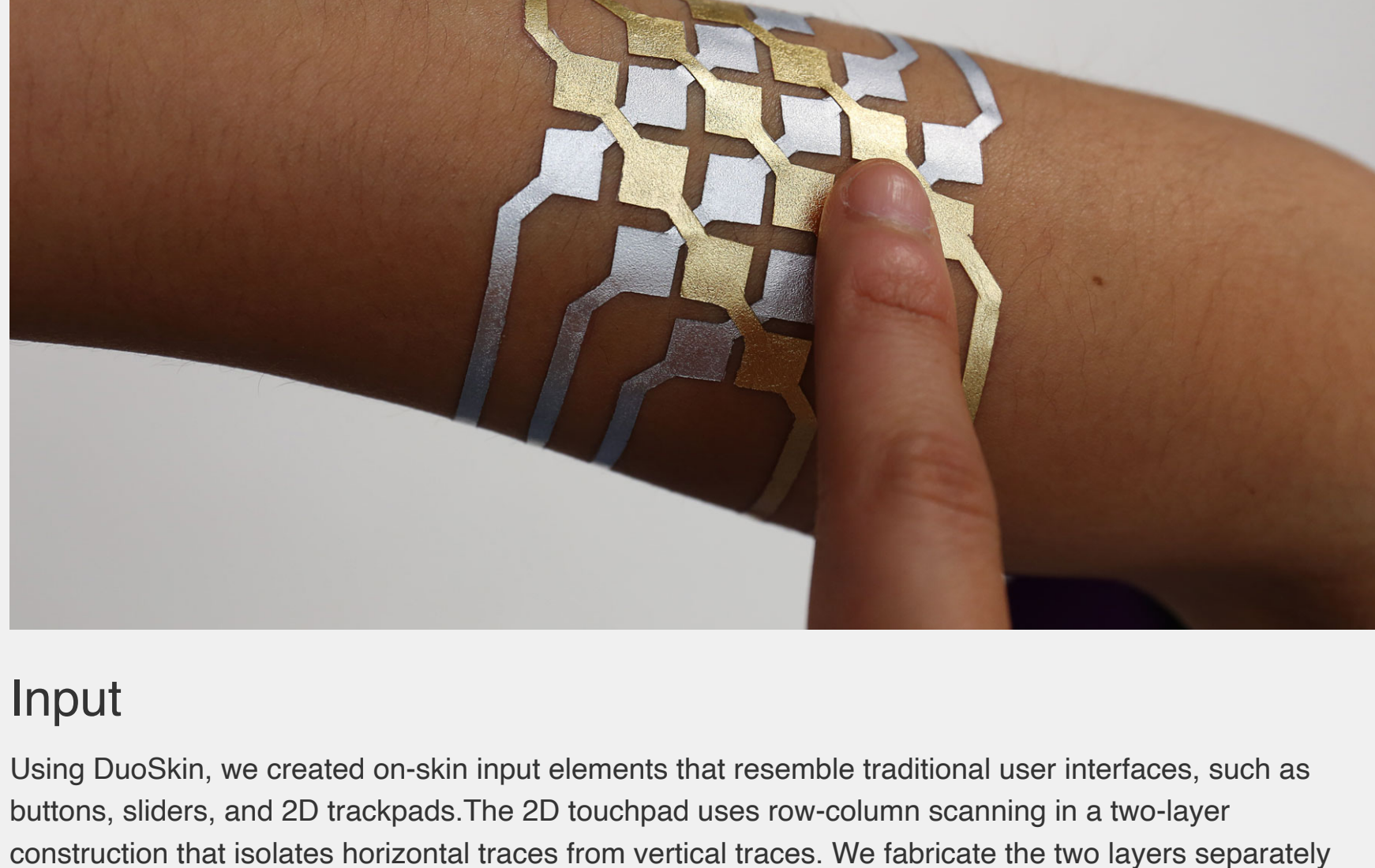
However, the manufacturing of epidermal electronics is expensive and requires materials and fabrication processes that are limited to the medical domain and material sciences. Indeed, applications commonly monitor vital signals [15], glucose levels [1], and skin conditions [16], but so far have not extended beyond health, activity, and wellness.

Researchers have sought to expand such on-skin devices from medical purposes to everyday user interaction. For example, iSkin is a flexible, visually customizable overlay that appropriates the skin for input [31]. Although promising, iSkin (700µm) is significantly thicker than existing epidermal electronics (0.8µm) [15] and still requires expert grade materials (silicone elastomer and carbon particles).

In this paper, we aim to make durable and skin-friendly *on-skin user interfaces* available to the wider community, using commodity materials, electronic components, and fabrication processes. We appropriate gold leaf as the key material to fabricate such wearable interactive devices, a type of metal leaf that appears golden. More than wearable around the body, the devices we showcase and their thin user interfaces attach directly to skin, bringing the benefits of epidermal electronics to low-cost interactive prototypes, which we call *DuoSkin* devices. The devices shown in Figure 1 sense input, display output, and communicate wirelessly with proximate devices. We created these devices exclusively using DuoSkin, whose process is outlined in Figure 2.

DuoSkin thus closes the gap of prior work on fabricating body interfaces [31] by repurposing accessible materials (e.g., gold leaf, thin tattoo paper) and tools (craft electronic

Three classes of on-skin user interfaces



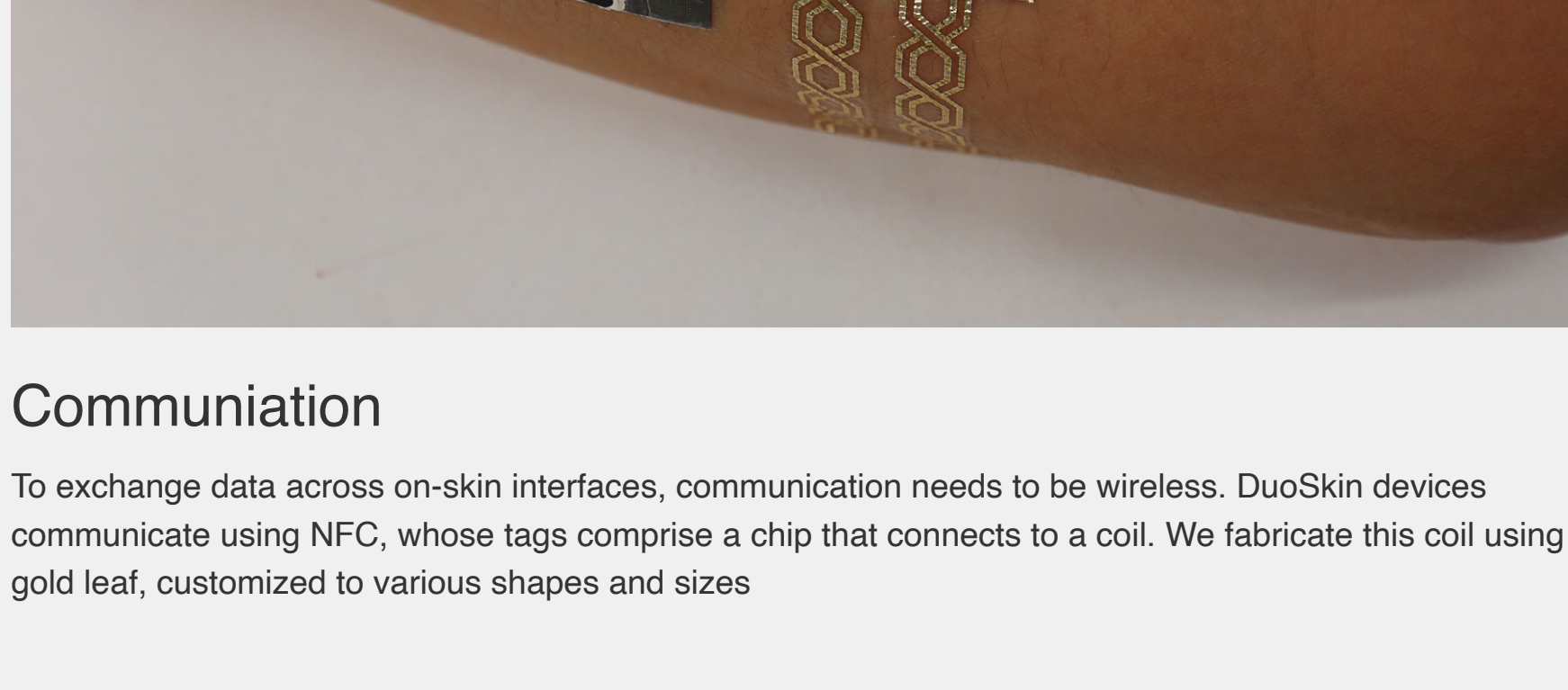
Input

Using DuoSkin, we created on-skin input elements that resemble traditional user interfaces, such as buttons, sliders, and 2D trackpads. The 2D touchpad uses row-column scanning in a two-layer construction that isolates horizontal traces from vertical traces. We fabricate the two layers separately and then apply and overlay them onto skin.



Output

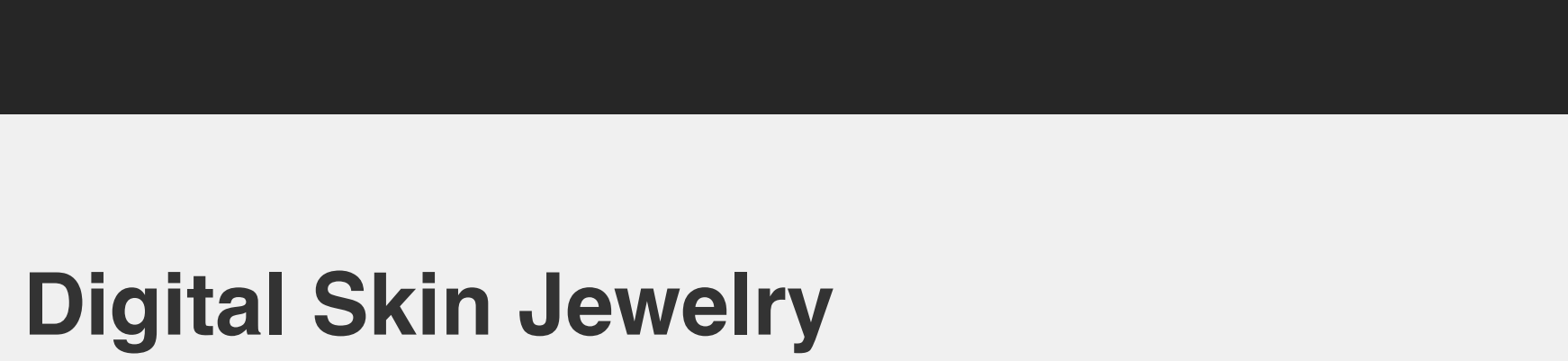
DuoSkin brings soft displays onto the skin, enabled through the ink-like qualities of thermochromic pigments. These displays have two different states and color change is triggered when heated beyond body temperature. Displays can also be separated into designated parts. To activate color changes on our displays, we fabricate resistive heating elements underneath the thermochromic layer.



Communication

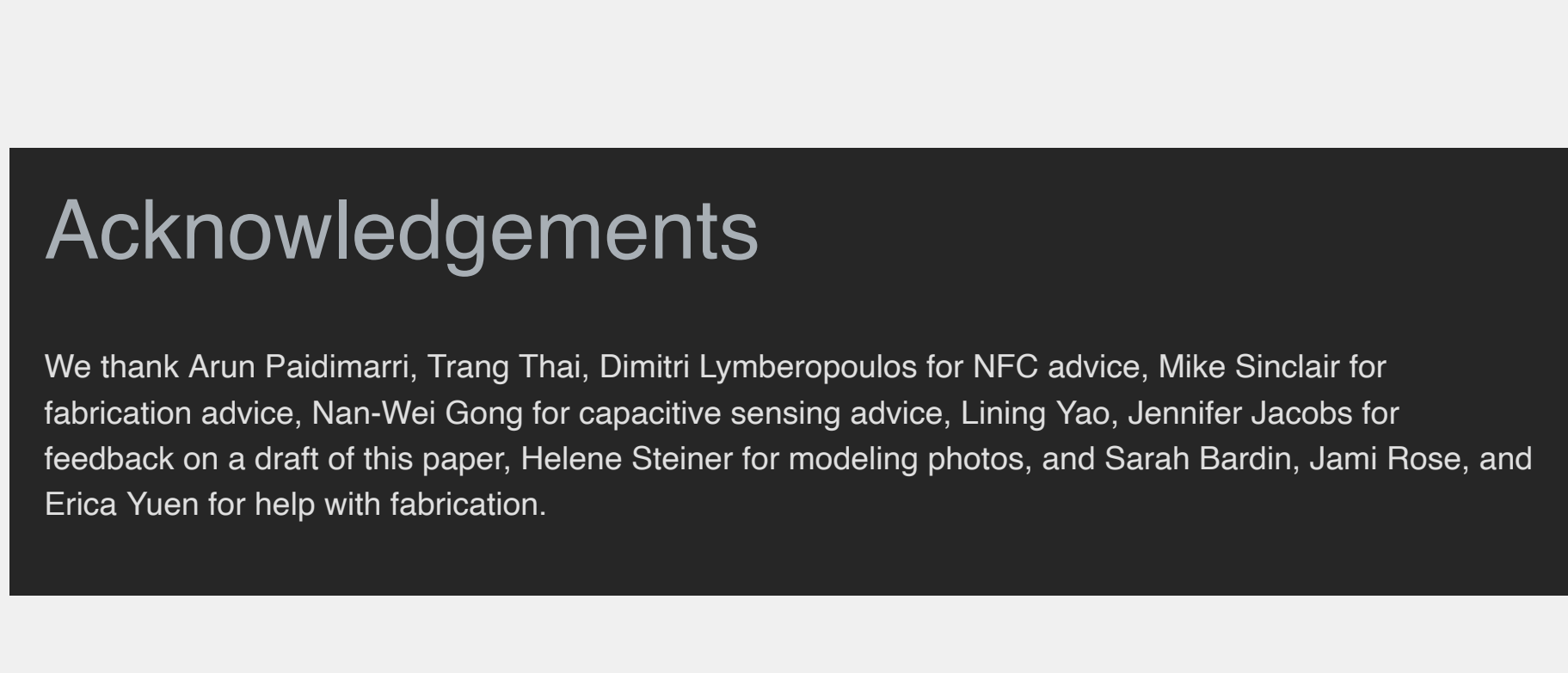
To exchange data across on-skin interfaces, communication needs to be wireless. DuoSkin devices communicate using NFC, whose tags comprise a chip that connects to a coil. We fabricate this coil using gold leaf, customized to various shapes and sizes

Fabrication Process



DuoSkin's three-step workflow. Step 1: (a) Sketching skin circuitry with graphic design software. Step 2: (b) Fabrication, which includes (c) creating stencils of the circuitry, (d) applying gold leaf as the conductive material, and (e) mounting electronics. Step 3: (f) After completing the circuitry, we apply the DuoSkin device to the user's skin through water-transfer.

Digital Skin Jewelry



DuoSkin metal leaf traces serve both, decorative and functional purposes, this LED necklace is an example of jewelry-like routing made with DuoSkin.

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